Check for updates

Hypertension

ORIGINAL ARTICLE

Finerenone Improves Albuminuria via MR-TRPC Signaling in Diabetic Kidney Disease

Tsukasa Iwakura, Kengo Kidokoro, Rie Tatsugawa, Akira Hirano, Eriko Kajimoto, Masanobu Takasu, Masafumi Wada, Yoshihisa Wada, Hiroyuki Kadoya, Seiji Kishi, Hajime Nagasu, David Z.I. Cherney, Tamaki Sasaki, Naoki Kashihara

BACKGROUND: Recent studies have confirmed the protective effects of nonsteroidal MR (mineralocorticoid receptor) antagonists in diabetic kidney disease. However, the physiological mechanisms underlying their albuminuria-reducing effects remain incompletely defined. We hypothesized that inhibition of the MR could protect podocytes by limiting excessive calcium influx via TRPC (transient receptor potential canonical) 5, thereby reducing albuminuria.

METHODS: We evaluated the effects of the nonsteroidal MR antagonist finerenone on albuminuria, podocyte morphology, and glomerular function in diabetic mice. Reactive oxygen species generation and single-nephron glomerular filtration rate were analyzed using in vivo imaging. Cultured podocytes were used to assess MR-TRPC5 signaling through measurements of Sgk1 (serum- and glucocorticoid-regulated kinase 1) and TRPC5 expression, intracellular calcium, and actin cytoskeletal organization.

RESULTS: Finerenone significantly reduced albuminuria, ameliorated podocyte morphological abnormalities, and decreased glomerular reactive oxygen species production in diabetic mice. In cultured podocytes, aldosterone increased Sgk1 and TRPC5 expression, elevated intracellular calcium, and induced actin reorganization; these changes were attenuated by finerenone and by the AC1903 (TRPC5 inhibitor). Activation of MR-TRPC signaling was associated with increased calcium influx and features of podocyte injury. In vivo imaging further indicated that finerenone was associated with lower single-nephron glomerular filtration rate, consistent with an attenuation of glomerular hyperfiltration.

CONCLUSIONS: Finerenone is associated with reductions in albuminuria in diabetic kidney disease, together with improvements in podocyte injury indices and glomerular hemodynamics. These benefits may be mediated in part through MR-TRPC5 signaling although additional pathways are likely to contribute. ***(Hypertension.* 2026;83:00–00. DOI: 10.1161/HYPERTENSIONAHA.125.25739.)** • **Supplement Material**.

Hypertension FIRST PROOF ONLY

Key Words: albuminuria ■ calcium signaling ■ diabetic nephropathies ■ glomerular filtration rate ■ podocytes

Downloaded from http://ahajournals.org by on February 11, 2026

Chronic kidney disease (CKD) affects 9.1% of the world's population and is increasing (Global Burden of Diseases, Injuries, and Risk Factors Study 2017). Fortunately, over the past 5 years, several effective and safe treatments for CKD, including diabetic kidney disease (DKD), have become available, such as SGLT (sodium-glucose cotransporter) 2 inhibitors, the nonsteroidal mineralocorticoid receptor antagonist (nsMRA) finerenone, and glucagon-like peptide-1 receptor agonists. Finerenone, one of the nsMRAs is kidney protective in patients with type 2 diabetes and CKD.[1–3] In addition, the clinical trials of finerenone are currently underway, targeting patients with CKD with type 1 diabetes, a population with limited treatment options, exploring its potential as a CKD treatment.[4] However, many aspects of the mechanisms by which mineralocorticoid receptor antagonist (MRA) inhibits the progression of renal injury remain unclear. Animal studies have reported that overactivation of the MR (mineralocorticoid receptor) in podocytes is involved in the progression of DKD

Correspondence to: Kengo Kidokoro, Department of Nephrology and Hypertension, Kawasaki Medical School, 577 Matsushima, Kurashiki, Okayama 701-0192, Japan. Email k.kid@med.kawasaki-m.ac.jp

Supplemental Material is available at https://www.ahajournals.org/doi/suppl/10.1161/HYPERTENSIONAHA.125.25739.
For Sources of Funding and Disclosures, see page XXX.

© 2026 American Heart Association, Inc.

Hypertension is available at www.ahajournals.org/journal/hyp

NOVELTY AND RELEVANCE

What Is New?

This study identifies podocyte calcium dynamics as a novel mechanistic target of finerenone in diabetic kidney disease.
Finerenone improved albuminuria and restored glomerular barrier integrity without affecting systemic hemodynamics.
Intravital multiphoton imaging visualized real-time podocyte calcium responses, revealing direct renoprotective actions of MR (mineralocorticoid receptor) antagonism.
The findings expand the mechanistic understanding of nonsteroidal MR blockade beyond conventional antifibrotic effects.

What Is Relevant?

Albuminuria reduction is a validated surrogate for renal outcome improvement in diabetic kidney disease, yet its cellular mechanisms remain incompletely understood.
Podocyte injury and cytoskeletal disorganization contribute critically to proteinuric progression.
Targeting podocyte MR signaling and calcium regulation may provide an additional therapeutic avenue for patients with persistent albuminuria despite optimized renin-angiotensin system inhibition.
These results complement ongoing clinical trials combining MR antagonists with SGLT2 (sodium-glucose cotransporter 2) inhibitors.

Clinical/Pathophysiological Implications?

Modulation of podocyte calcium homeostasis represents a novel pathophysiological axis that links MR activation to glomerular injury.
Pharmacological MR inhibition by finerenone may directly stabilize podocyte structure and function, contributing to durable albuminuria regression.
Integrating mechanistic imaging biomarkers with MR-targeted therapy could accelerate precision medicine approaches in diabetic kidney disease.
These insights support further translational and clinical studies examining podocyte-specific MR signaling and calcium-dependent pathways as determinants of kidney protection.

Hypertension

Nonstandard Abbreviations and Acronyms

AC1903	transient receptor potential canonical 5 inhibitor
BP	blood pressure
BSA	bovine serum albumin
CKD	chronic kidney disease
DKD	diabetic kidney disease
EGFP	enhanced green fluorescent protein
GCaMP5	genetically encoded calcium indicator 5
GFP	green fluorescent protein
GFR	glomerular filtration rate
HS	high salt
LY	Lucifer yellow
MCP-1	monocyte chemoattractant protein-1
MMP-2	matrix metalloproteinase-2
MR	mineralocorticoid receptor
MRA	mineralocorticoid receptor antagonist
NKCC2	Na-K-2Cl cotransporter 2
nNOS (NOS1)	neuronal nitric oxide synthase
nsMRA	nonsteroidal mineralocorticoid receptor antagonist
PAI-1	plasminogen activator inhibitor-1
ROS	reactive oxygen species
SBP	systolic blood pressure
Sgk1	serum- and glucocorticoid-regulated kinase 1
SGLT	sodium-glucose cotransporter
SNGFR	single-nephron glomerular filtration rate
tdTomato	tandem dimer Tomato fluorescent protein
TIMP-1	tissue inhibitor of metalloproteinases-1
TRPC	transient receptor potential canonical
UNx	uninephrectomy
UNx-HS	uninephrectomy plus high-salt diet

and albuminuria, and is also linked with hypertension, heart failure, and metabolic abnormalities, including type 2 diabetes.[5]

From a mechanistic perspective, TRPC (transient receptor potential canonical) channels, specifically TRPC5 and TRPC6, have been associated with increased Ca^{2+} influx into podocytes, leading to cytoskeletal remodeling and subsequent albuminuria.[6–8] In experiments using coronary arteries from animals with

Downloaded from http://ahajournals.org by on February 11, 2026

metabolic syndrome, the steroidal MRA spironolactone reduced protein expression of TRPC1 and TRPC6 and inhibited atherosclerosis,[9] suggesting that MR activation may be involved in the regulation of TRPC expression. However, there are no reports investigating MR activation and TRPC expression in podocytes. In addition, TRPC5 and TRPC6 activity is augmented by oxidative stress. Because MR activation enhances reactive oxygen species (ROS) production via nongenomic pathways,[10] MRAs have the potential to inhibit both TRPC expression and activity.

From a hemodynamic perspective, in clinical trials with finereone, there was an initial dip in estimated glomerular filtration rate (GFR) 2 weeks after nsMRA administration, followed by a slower decline compared with placebo.[1] This pattern is similar to estimated GFR changes observed in clinical trials of renin-angiotensin system inhibitors and SGLT2 inhibitors, suggesting that nsMRA may reduce glomerular hyperfiltration. Thus, beyond nonhemodynamic pathways such as TRPC, nsMRAs may also regulate GFR, potentially contributing to their renoprotective effects in DKD.

Accordingly, the aim of this study was to elucidate the mechanisms responsible for kidney protection with the nsMRA finerenone. We hypothesized that activation of the MR-TRPC pathway in podocytes and glomerular hyperfiltration due to MR activation is involved in the progression of renal damage in DKD, and administration of finerenone would attenuate these deleterious pathways.

Hypertension FIRST PROOF ONLY

METHODS

Data Availability

All summary data supporting the findings of this study are contained in the article and its Supplemental Material. Deidentified raw and processed data sets underlying the figures, the analysis code, and detailed experimental materials and protocols are available from the corresponding author upon reasonable request and may require institutional approvals and data use or material transfer agreements where applicable. The corresponding author will maintain continued availability.

Experimental Materials

The experimental protocol (approval number: 23-063, approval date: May 1, 2023) was approved by the Ethics Review Committee for Animal Experimentation at Kawasaki Medical School, Kurashiki, Japan. C57BL/6 (control; C57BL/6JmsSlc, Japan SLC, Shizuoka, Japan; RRID: IMSR_SLC:B6) and *Ins2*$^{+/Akita}$ mice (Japan SLC; RRID: IMSR_JAX:003548) were purchased from Japan SLC. Podocin-GCaMP5 (genetically encoded calcium indicator 5)/tdTomato (tandem dimer Tomato fluorescent protein) mice were created by crossing podocyte-specific Cre recombinant (podocin-cre) mice with GCaMP5/tdTomato flox mice. The podocin-cre mice, originally obtained from The Jackson Laboratory (RRID: IMSR_JAX:008205, Bar Harbor, ME), were provided by Dr Kojiro Nagai from Tokushima University through a material transfer agreement. GCaMP5/tdTomato flox mice were purchased from The Jackson Laboratory (RRID: IMSR_JAX:024477). GCaMP is a calcium indicator created from a fusion of EGFP (enhanced green fluorescent protein), calmodulin, and M13, a peptide sequence derived from myosin light-chain kinase. All mice were backcrossed ≥10 generations to C57BL/6JmsSlc mice before being used for experiments. The mice were housed in a temperature- and humidity-controlled room under a 12-hour light-dark cycle. They were had free access to tap water. Uninephrectomized 7 weeks old C57BL/6 (uninephrectomy [UNx]-control) and *Ins2*$^{+/Akita}$ mice were fed normal salt (0.5% NaCl) or high salt (HS; 3% NaCl; UNx plus HS diet [UNx-HS] control, UNx-HS *Ins2*$^{+/Akita}$) diet for 8 weeks. Uninephrectomized *Ins2*$^{+/Akita}$ group was fed a diet with the nonsteroidal MRA finerenone (BAYER AG, Wuppertal, Germany) orally at a dose of 10 mg/kg per d for 8 weeks (UNx-HS *Ins2*$^{+/Akita}$+FINE; Figure S1A). Streptozotocin was administered intraperitoneally at a dose of 50 mg/kg for 5 consecutive days to induce diabetes in 6-week-old podocin-GCaMP5/tdTomato mice. Animals were considered to be in a diabetic state if their fasting glucose levels 1 week after streptozotocin administration were ≥250 mg/dL. The mice underwent unilateral nephrectomy at 8 weeks of age, and drug treatment and HS diet were initiated at 9 weeks of age for 6 weeks. Finerenone was administered orally at a dose of 10 mg/kg per d, and AC1903 (TRPC5 inhibitor; HY-122051, MedChemExpress, Monmouth Junction, NJ) was administered intraperitoneally at a dose of 50 mg/kg per d for 6 weeks (Figure S1B). The UNx was performed under experienced personnel's guidance. Before surgery, nociceptive reflex absence was confirmed via hindlimb pinch under inhalation anesthesia. Pain responses were continuously monitored. The surgical field remained sterile. After right kidney removal, penicillin G (20 000 units/kg, IP) was given for 3 days. Buprenorphine (0.05 mg/kg, SC) was administered every 12 hours for 3 days for pain relief. Infection cases had a contingency for additional Penicillin G (20 000 units/kg, IM, every 24 hours) though it was unnecessary. Euthanasia was planned but not required. Only male *Ins*$^{2+/Akita}$ mice were used due to the lower blood glucose levels observed in females. Stratified randomization was performed. To ensure allocation concealment, animals were numbered at birth by one investigator, and group assignment was independently conducted by another investigator who was blinded to the numbering. Blood pressure (BP) and pulse rate were measured by the tail-cuff method using an automatic sphygmomanometer (BP98A; Softron, Tokyo, Japan, Supplemental Method S1). Twenty-four-hour urine samples were collected using metabolic cages. Urinary albumin levels were determined by ELISA with a murine microalbuminuria ELISA kit (Albuwell M; Exocell).

Multiphoton In Vivo Imaging

The in vivo imaging procedure was performed as described previously.[11] Alexa Fluor 594–conjugated BSA (A13101), fluorescein isothiocyanate–conjugated BSA (A23015), and Lucifer yellow (LY; L453) were purchased from Molecular Probes, Inc (Eugene, OR). Alexa Fluor 594–conjugated BSA (bovine serum albumin) were used to label the plasma flow (100 µL of 5-mg/mL stock; Invitrogen, Carlsbad, CA). Fluorescein isothiocyanate–conjugated BSA was used to assess the glomerular permeability of albumin (100 µL of 5-mg/mL stock; Invitrogen).

Downloaded from http://ahajournals.org by on February 11, 2026

LY was injected as a bolus to measure single-nephron GFR (SNGFR; 10 µL of 10-mg/mL stock; Invitrogen).

Measurement of SNGFR

SNGFR was measured using a previously described procedure.[11,12] Alexa Fluor 594–conjugated BSA was injected through the catheter to visualize blood flow (10 µL of a 5 mg/mL). After identifying a superficial glomerulus and subsequent proximal tubule that could be observed over a minimum length of 100 µm, LY was injected into the carotid artery, and the transit of the filtered LY was followed downstream in the proximal tubule. SNGFR was calculated by measuring the fluorescence intensity change associated with LY in 2 regions of interest ≈100 to 200 µm apart in the early proximal tubule segment, the internal diameter, and the length of the tubule: SNGFR=(volume)/(time), where (volume)=(length×[internal diameter/2]2×ϖ) and (time)=the LY transit time between the first and second regions of interest, which was calculated at the peak of LY fluorescence intensity. Five-to-10 image series (xyt, 60 Hz) were collected for 3 to 5 mice per group. The detailed experimental procedures are described in Supplemental Method S2.

In Situ Detection of NO and ROS

The production of NO and ROS was assessed by confocal laser microscopy after renal perfusion with 4-amino-5-methylamino-2',7'-difluorofluorescein diacetate (NO probe; excitation/emission maxima, 495/515 nm; GORYO Chemical, Inc, Sapporo, Japan) and CellROX Deep Red (excitation/emission maxima, 665/644 nm; C10422, RRID: AB_2556216, Invitrogen, Thermo Fisher Scientific K. K., Tokyo, Japan), as previously described.[13] Quantification was performed on glomerular tuft regions of interest using background-subtracted 3-dimensional mean fluorescence intensity. Detailed procedures for in situ detection, image acquisition, and quantification are provided in Figure S2.

Western Blotting

Cell lysates were extracted with extraction buffer or sample buffer, as described previously.[14] Protein samples were subjected to immunoblotting analysis with antibodies against TRPC5 (25890-1-AP; RRID: AB_11142010, Proteintech, Rosemont, IL) and Sgk1 (serum- and glucocorticoid-regulated kinase 1; ab32374, RRID: AB_776654, Abcam, Cambridge, United Kingdom). Each Western blot was performed using samples from 4 to 6 mice per group. Detailed information about Western immunoblotting is provided in Supplemental Method S6.

Cell Culture

A conditionally immortalized mouse podocyte cell line was kindly provided by Katsuhiko Asanuma (Chiba University Hospital, Chiba, Japan).[15] This cell line was cultured using the previously described method.[16]

Confocal Calcium Imaging in Cultured Podocytes

Calcium imaging of cultured podocytes was performed as previously described.[16] Differentiated podocytes were seeded into 24-well plates and loaded with calcium-sensitive dyes Fluo-4-AM (F311, λ_ex=495 nm, λ_em=518 nm; F311, RRID: AB_2665512, Dojindo, Kumamoto, Japan) and Fura Red-AM (F3021, λ_ex=436/473 nm, λ_em=655/670 nm; F3021, RRID: AB_2665513, Invitrogen/Molecular Probes, Inc). Images were acquired using a Nikon A1R-MP multiphoton confocal microscope equipped with an inverted imaging system and an Apo LWD 25×/1.10 W DIC N2 objective lens, at 3 s per frame using Nikon imaging software NIS Elements Advanced Research (RRID: SCR_014329). After dye loading, aldosterone (10 µL; final concentration=100 nmol/L) was added to wells, and the effects on the podocytes were examined. The incubation times and concentration of agents used are given as follows: finerenone: 100 nM/24 h and AC1903: 30 µM/3 h. The changes in intracellular Ca^{2+} concentration were estimated using the ratio of Fluo-4-AM to Fura Red-AM fluorescence intensity. Detailed information about calcium imaging of cultured podocytes is provided in Supplemental Method S3.

Detection of ROS in Cultured Podocytes

ROS production was detected by 2',7'-dichlorofluorescein staining.[17] Fluorescence images were obtained using a Nikon A1R-MP multiphoton confocal microscope equipped with an inverted imaging system at excitation/emission wavelengths of 485/535 nm for 2',7'-dichlorofluorescein. The fluorescence intensity values from 15 to 25 different cultured podocytes were calculated. The incubation times and concentration of agents used are given as follows: finerenone: 100 nM/24 h and aldosterone: 100 nM/3 h.

Immunocytochemistry

For F-actin staining, cells were stained with Alexa Fluor 488 phalloidin (A12379, RRID: AB_2315147, Invitrogen) for 20 minutes at 20 °C. After several washes, the nuclei were labeled with Hoechst 33342 solution (dilution 1/5000, H3570, RRID: AB_2629482, Invitrogen) for 30 s.

Confocal Calcium Imaging in the Podocytes of Isolated Glomeruli

Calcium imaging of isolated glomeruli was performed as previously described.[16] Then, aldosterone (10 µL; final concentration, 50 nmol/L) or hydrogen peroxide (H_2O_2, 10 µL; final concentration, 50 nmol/L) was added to the wells, and the glomeruli were monitored for evaluation of GFP (green fluorescent protein) intensity changes. Detailed experimental conditions and analytical procedures are described in Supplemental Method S4.

Podocytes Calcium Imaging in vivo

Podocyte Ca^{2+} status in diabetic podocin-GCaMP5/tdTomato mice was visualized using multiphoton microscopy. In vivo podocyte GCaMP5 fluorescence intensity was quantified as an index of steady-state intracellular Ca^{2+} levels under identical imaging conditions. Detailed quantification and normalization procedures are described in Supplemental Method S5.

Statistical Analysis

All values are expressed as mean±SEM. Statistical analyses were performed using GraphPad Prism10 software (RRID:

Downloaded from http://ahajournals.org by on February 11, 2026

Hypertension FIRST PROOF ONLY

SCR_002798, GraphPad Software, La Jolla, CA). Comparisons between 2 groups were performed using unpaired or paired 2-tailed Student *t* tests or Mann-Whitney *U* tests. Comparisons between multiple groups that comprised normally distributed data were performed using 1-way ANOVA, followed by the Tukey multiple comparisons test. Comparisons between multiple groups comprising nonnormally distributed data were performed using the Kruskal-Wallis test, followed by the Dunn multiple comparisons test. Statistical significance was set at $P<0.05$.

RESULTS

Effects of Finerenone on Glomerular Injury and Albuminuria in Type 1 Diabetes Model Mice

The physiological and serum sample data for each group are summarized in Tables S1 and S2. There were no differences in physiological parameters between the 2 diabetic groups. Within each background (control or diabetic), body weight did not differ between normal-salt and HS diets (controls: 26.5±0.6 versus 26.7±0.8 g; diabetics: 24.0±0.3 versus 23.1±0.4 g). Daytime, resting tail-cuff systolic BP (SBP) tended to be lower with HS in controls (UNx-HS-control: 104.7±2.1 versus UNx-control: 111.8±1.7 mm Hg) and was lower in the diabetic cohorts than in controls (*Ins2*$^{+/Akita}$: 99.5±1.9; *Ins2*$^{+/Akita}$+FINE: 101.3±3.4 mm Hg; $P<0.05$ versus UNx-control). Importantly, the finerenone regimen used in this study did not produce a significant antihypertensive effect, as SBP in *Ins2*$^{+/Akita}$+FINE was not lower than in *Ins2*$^{+/Akita}$. Longitudinal SBP values at 0, 4, and 8 weeks (mean of 3 consecutive readings at each time point) are provided in Figure S3. Histological evaluation showed that UNx-HS *Ins2*$^{+/Akita}$ mice had increased mesangial matrix by periodic acid-Schiff staining, which was partially suppressed by finerenone (Figure 1A and 1B). Electron microscopy revealed podocyte foot process effacement in UNx-HS *Ins2*$^{+/Akita}$ mice, which was attenuated by finerenone (Figure 1C). To quantify effacement, we measured foot process width along the glomerular basement membrane; foot process width was increased in UNx-HS *Ins2*$^{+/Akita}$ mice and was reduced by finerenone (Figure 1D). Urinary albumin excretion was significantly increased in UNx-HS *Ins2*$^{+/Akita}$ mice compared with UNx-control (UNx-control: 11.4±0.9 µg/d versus UNx-HS *Ins2*$^{+/Akita}$: 219.6±25.5 µg/d; $P<0.05$; Figure 1E) and

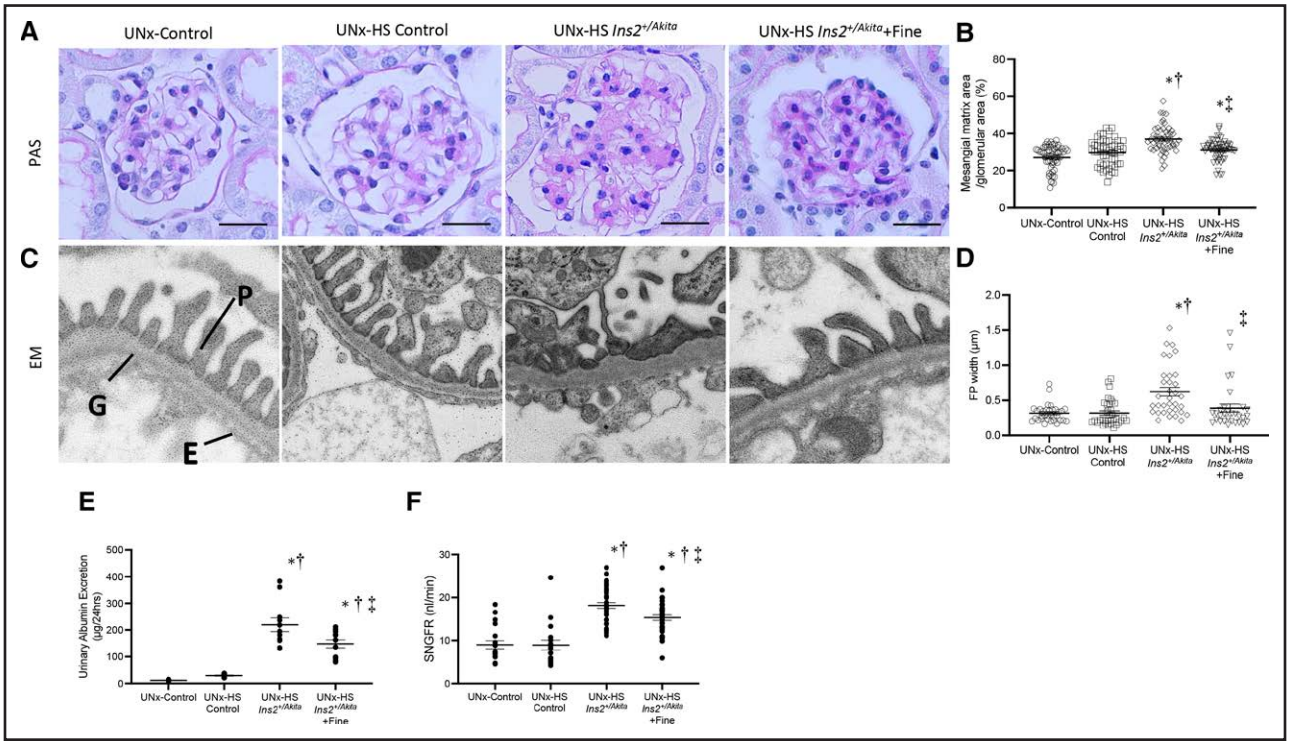

Figure 1. Effect of finerenone on glomerular injury, albuminuria, and single-nephron glomerular filtration rate (SNGFR) in diabetic *Ins2*$^{+/Akita}$ mice.

A, Representative periodic acid-Schiff (PAS) staining images of renal sections. **B**, Ratio of mesangial area to total glomerular area; analyses were performed in a blinded manner with 6 mice per group and 8 to 10 glomeruli per mouse (total 50–55 glomeruli per group). **C**, Electron microscopy images. **D**, Quantification of podocyte foot process (FP) width on transmission electron microscopy (TEM) micrographs; analyses were blinded, with 3 mice per group, 1 glomerulus per mouse, and 2 fields per glomerulus, yielding a total of ≈30 to 35 FPs measured per group. **E**, Urinary albumin excretion for 24 hours. **F**, SNGFR measurements (19 glomeruli in uninephrectomy [UNx]-control [n=3 mice], 18 glomeruli in UNx plus high-salt diet [UNx-HS] control [n=3 mice], 38 glomeruli in UNx-HS *Ins2*$^{+/Akita}$ [n=5 mice], and 38 glomeruli in UNx-HS *Ins2*$^{+/Akita}$+finerenone [n=5 mice]). Mean±SEM. Scale bar, 40 µm. *$P<0.05$ vs UNx-control; †$P<0.05$ vs UNx-HS-control; and ‡$P<0.05$ vs UNx-HS *Ins2*$^{+/Akita}$.

Downloaded from http://ahajournals.org by on February 11, 2026

was partially reduced with finerenone (UNx-HS *Ins2*$^{+/Akita}$ versus UNx-HS *Ins2*$^{+/Akita}$+finerenone: 148.0±15.3 µg/d; $P<0.05$; Figure 1E), yet remained higher than in UNx-control. UNx-HS *Ins2*$^{+/Akita}$ mice exhibited a significantly higher SNGFR compared with UNx-control mice (UNx-control: 8.98±0.94 nL/min versus UNx-HS *Ins2*$^{+/Akita}$: 18.16±0.66 nL/min; $P<0.05$; Figure 1F). Finerenone significantly lowered SNGFR in UNx-HS *Ins2*$^{+/Akita}$ mice, but values did not normalize to UNx-control (UNx-HS *Ins2*$^{+/Akita}$ versus UNx-HS *Ins2*$^{+/Akita}$+finerenone: 15.32±0.66 nL/min; $P=0.02$; Figure 1F).

Finerenone Suppressed Glomerular ROS Production and Protein Expression of Sgk1 and TRPC5 in Mouse Kidney

Glomerular ROS and NO signals were quantified by confocal Z-stack imaging using CellROX Deep Red and 4-amino-5-methylamino-2',7'-difluorofluorescein diacetate (NO probe) and expressed as background-subtracted 3-dimensional mean fluorescence intensity. Compared with controls, *Ins2*$^{+/Akita}$ mice exhibited a significant increase in ROS 3D MFI and a significant decrease in NO 3D MFI. Finerenone treatment significantly reduced ROS 3D MFI compared with *Ins2*$^{+/Akita}$ although ROS levels remained higher than controls (Figure 2A and 2B). We next examined TRPC5 and Sgk1 protein expression in the whole kidney and isolated glomeruli. In whole kidney samples, Sgk1, a downstream target of MR, and TRPC5 were both upregulated in *Ins2*$^{+/Akita}$ mice, indicating MR activation. These increases were attenuated by finerenone treatment (Figure 2C through 2E). Similarly, in isolated glomeruli, the protein expressions of TRPC5 and Sgk1 were markedly increased in *Ins2*$^{+/Akita}$ mice. Finerenone treatment significantly suppressed the expression of both proteins (Figure 2F through 2H).

Inhibitory Effects of Finerenone and AC1903 on Aldosterone-Induced Calcium Influx and Cellular Changes in Cultured Mouse Podocytes

Cultured mouse podocytes were used to investigate the effects of aldosterone stimulation on intracellular calcium influx, as well as the inhibitory effects of finerenone and the TRPC5 inhibitor, AC1903. In cultured mouse podocytes stimulated with aldosterone, a rapid increase in Ca^{2+} concentration was observed in the control group. However, in podocytes pretreated with finerenone or AC1903, this increase in Ca^{2+} concentration was suppressed (Figure 3A and 3B). Notably, the inhibitory effect appeared more pronounced with finerenone than with AC1903. In addition, TRPC5 protein expression was assessed by Western blotting in cultured podocytes after 24-hour aldosterone treatment. Aldosterone induced a dose-dependent increase in TRPC5 expression (Figure 3C). Furthermore, cotreatment with finerenone significantly suppressed TRPC5 expression compared with aldosterone treatment alone (Figure 3D). Evaluation of ROS production in cultured podocytes using 2',7'-dichlorodihydrofluorescein diacetate staining showed increased ROS production in aldosterone-stimulated podocytes, which was suppressed by finerenone (Figure 3E). Phalloidin staining was then performed to assess changes in the actin cytoskeleton due to calcium influx in podocytes. In the control group, F-actin was uniformly distributed, whereas aldosterone or H_2O_2-stimulated podocytes exhibited actin rearrangement. This actin rearrangement was suppressed by finerenone and AC1903 (Figure 3F). In addition, the ρ family of small GTPases plays an important role in the organization of podocyte foot process structure (28), and actin filaments are important for the maintenance of normal podocyte morphology and motility (29). Rac1 activity, one of the ρ GTPases, was elevated by aldosterone stimulation and showed a tendency to be suppressed by finerenone, with significant suppression observed with AC1903 (Figure 3G).

Evaluation of Calcium Influx Into Podocytes in Isolated Glomeruli From Diabetic Mice, Following Aldosterone or H_2O_2 Stimulation

Next, we evaluated the calcium influx into podocytes in isolated glomeruli from podocin-GCaMP5/tdTomato mice induced with diabetes using streptozotocin, followed by UNx and HS diet, with or without finerenone, following aldosterone or H_2O_2 stimulation. The data on BP, body weight, and blood glucose levels in mice are provided in the Supplemental Material (Table S3). Both aldosterone and H_2O_2 stimulation in streptozotocin diabetic mice induced calcium influx into podocytes. Aldosterone stimulation took an average of ≈8 minutes to confirm calcium influx (Figure 4A and 4B), whereas H_2O_2-induced calcium influx was observed immediately after stimulation (Figure 4C and 4D). Finerenone treatment suppressed the calcium influx in response to both stimuli (Figure 4A through 4D).

Evaluation of Calcium Concentration Changes in Podocytes of Diabetic Mice In Vivo and the Effects of Finerenone and AC1903 Treatment

Changes in podocyte calcium in vivo using streptozotocin-induced diabetic podocin-GCaMP5/tdTomato mice were also evaluated. In streptozotocin-induced diabetic podocin-GCaMP5/tdTomato mice, an increase in podocyte calcium concentration was observed, which was significantly suppressed with finerenone (Figure 5B). AC1903 also suppressed the increase in podocyte calcium concentration, but, when combined with finerenone, the decrease in calcium concentration was more pronounced

Hypertension FIRST PROOF ONLY

Downloaded from http://ahajournals.org by on February 11, 2026

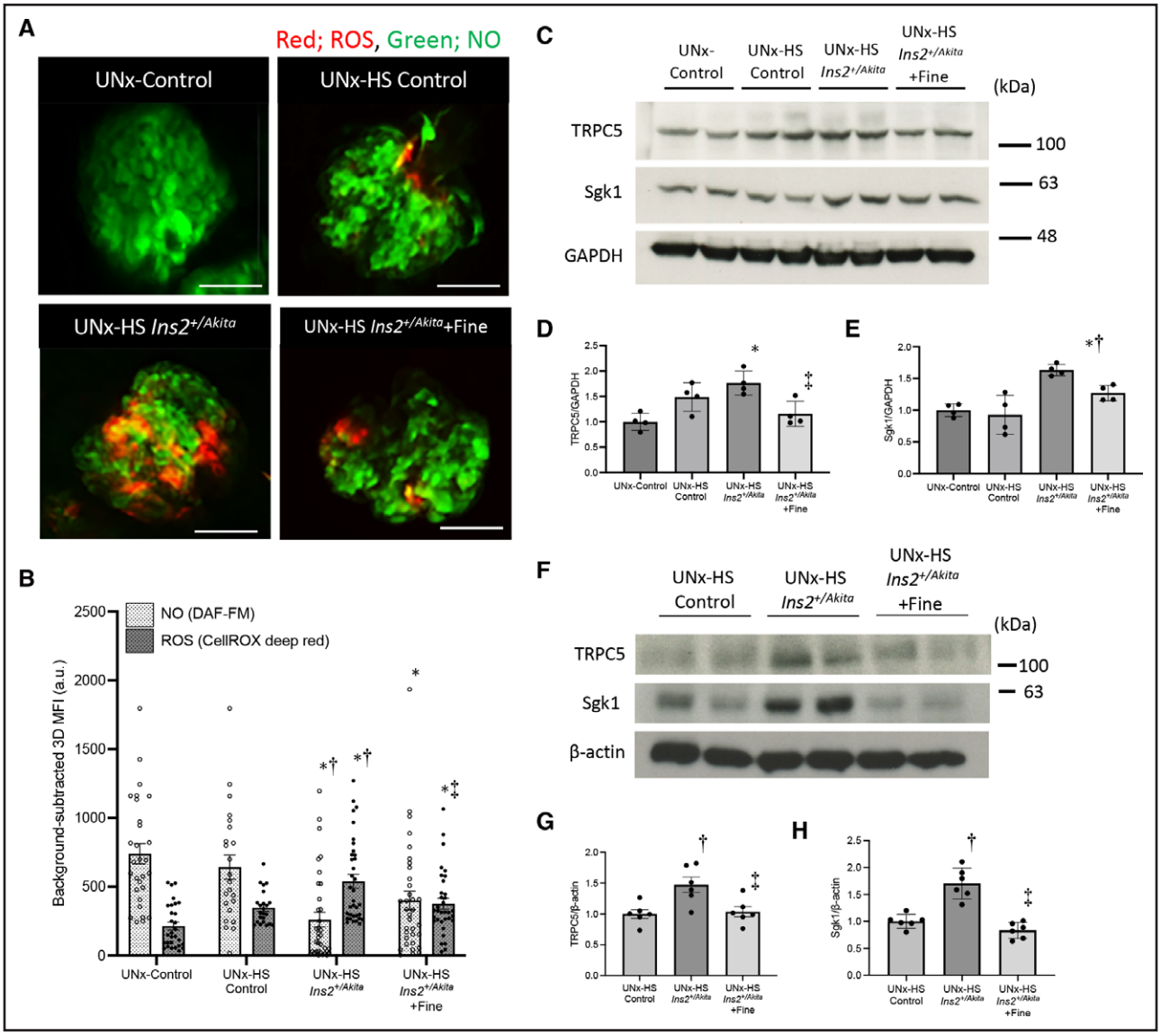

Figure 2. In situ detection of reactive oxygen species (ROS) and NO signals and protein expression of TRPC5 (transient receptor potential canonical 5) and Sgk1 (serum- and glucocorticoid-regulated kinase 1) in the kidneys of diabetic *Ins2*$^{+/Akita}$ mice.
A, Representative confocal images of glomerular ROS (CellROX Deep Red; red) and NO (4-amino-5-methylamino-2',7'-difluorofluorescein diacetate [NO probe] DAF-FM DA]; green) in the renal cortex. **B**, Quantification of glomerular ROS and NO signals expressed as background-subtracted 3D mean fluorescence intensity obtained from confocal Z-stacks. n=3 mice per group; 23 to 35 glomeruli analyzed per group (7–12 glomeruli per mouse; 2–3 sections per kidney). **C**, Western blotting analysis of TRPC5 and Sgk1 in whole kidney. **D** and **E**, Quantification of the Western blotting in whole kidney (n=4). **F**, Western blotting analysis of TRPC5 (n=6) and Sgk1(n=6) in isolated glomeruli. **G** and **H**, Quantification of the Western blotting in isolated glomeruli. Mean±SEM. Scale bar, 50 μm. *P<0.05 vs uninephrectomy (UNx)-control; †P<0.05 vs UNx plus high-salt diet (UNx-HS)-control; and ‡P<0.05 vs UNx-HS *Ins2*$^{+/Akita}$.

(Figure 5C). In addition, in the 3 groups shown in Figure 5C, we measured 24-hour urinary albumin excretion and SNGFR. AC1903 monotherapy significantly reduced albuminuria (UNx-HS/streptozotocin: 70.7±3.1 μg/d versus UNx-HS/streptozotocin+AC1903: 53.8±5.3 μg/d; P=0.043) but did not materially alter SNGFR versus diabetic controls (UNx-HS/streptozotocin: 18.2±1.8 nL/min versus UNx-HS/streptozotocin+AC1903: 16.2±1.1 nL/min; P=0.512). By contrast, the AC1903+finerenone combination significantly decreased both albuminuria (UNx-HS/streptozotocin+AC1903+FINE: 43.9±4.5 μg/d; P=0.002) and SNGFR (11.4±0.9 nL/min; P=0.002).

Assessment of Aldosterone-Induced Calcium Influx Into Podocytes in Isolated Glomeruli From Diabetic Mice Treated With AC1903 and Finerenone

In addition, in isolated glomeruli from streptozotocin mice, streptozotocin mice treated with AC1903, and streptozotocin mice treated with AC1903 and finerenone, we

Downloaded from http://ahajournals.org by on February 11, 2026

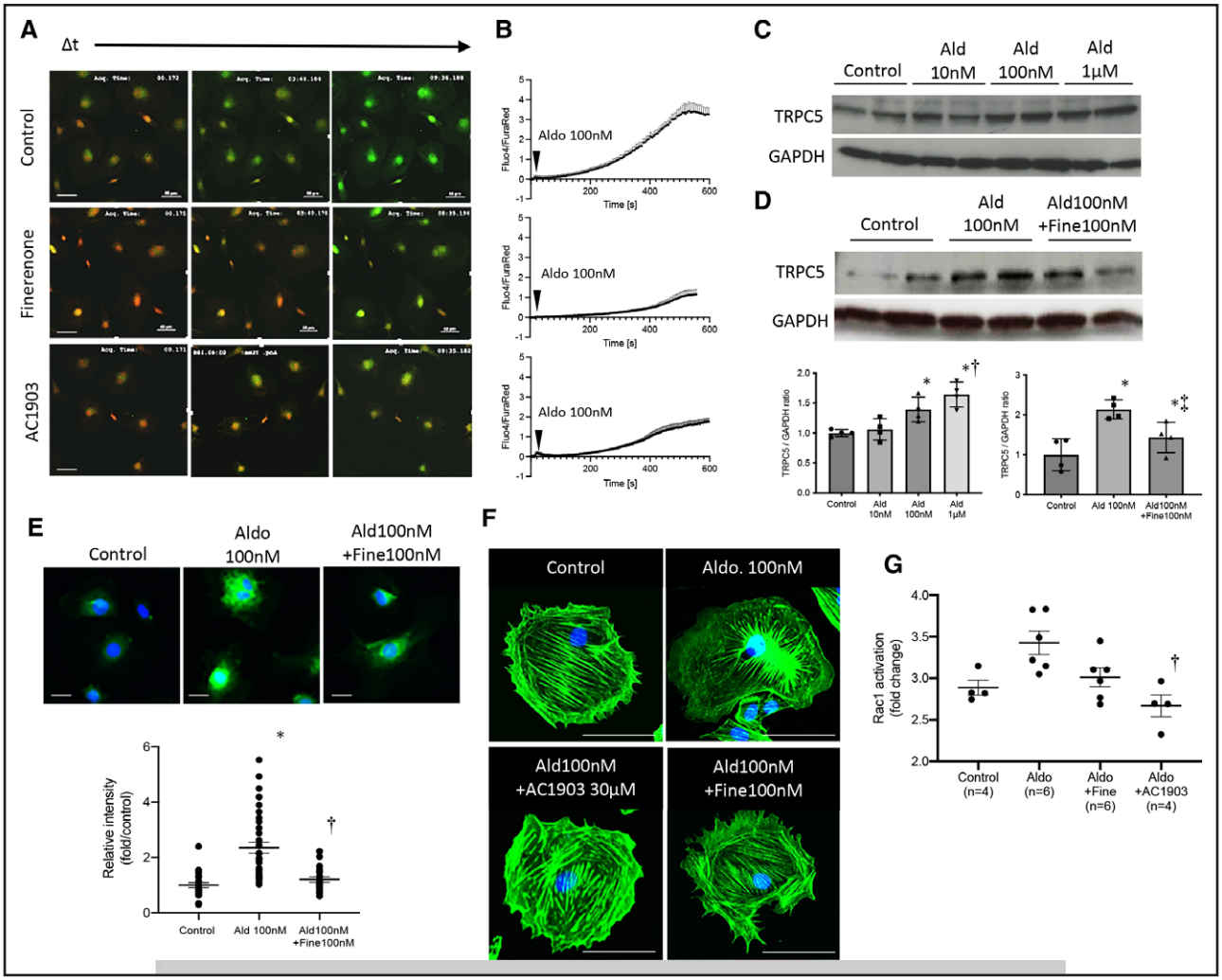

Figure 3. Inhibitory effects of finerenone and AC1903 (transient receptor potential canonical 5 inhibitor) on aldosterone-induced calcium influx and actin rearrangement in cultured mouse podocytes.

A and **B**, Ca^{2+} influx into cultured mouse podocytes under aldosterone stimulation, with and without pretreatment of finerenone or AC1903. **C**, Dose-dependent changes in TRPC5 (transient receptor potential canonical 5) protein expression in cultured mouse podocytes following aldosterone stimulation. **D**, Inhibition of TRPC5 expression by finerenone in cultured mouse podocytes. **E**, Reactive oxygen species (ROS) production in cultured mouse podocytes assessed by 2',7'-dichlorodihydrofluorescein diacetate (DCFH-DA) staining after aldosterone stimulation, with and without finerenone treatment. **F**, Actin cytoskeleton distribution in cultured mouse podocytes visualized by phalloidin staining, following stimulation with aldosterone with finerenone or AC1903. **G**, Rac1 activity in cultured mouse podocytes after aldosterone stimulation, with finerenone or AC1903 treatment. Mean±SEM. Scale bar, 50 μm. *P<0.05 vs control; †P<0.05 vs Ald 10 nM; and ‡P<0.05 vs Ald 100 nM.

FIRST PROOF ONLY

assessed calcium influx into podocytes stimulated by aldosterone. In isolated glomeruli from diabetic mice, aldosterone stimulation induced a significant calcium influx (Figure 6A). However, in isolated glomeruli from mice treated with AC1903 or finerenone+AC1903, the calcium influx into podocytes was markedly suppressed (Figure 6B and 6C).

DISCUSSION

The aim of this study was to elucidate the molecular mechanisms underlying kidney protection with nsMRA in DKD. Our hypothesis was that inhibition of MR activity could protect podocytes by limiting excessive calcium influx through TRPC5, thereby reducing albuminuria. Finerenone reduced albuminuria in diabetic mice, along with alleviating morphological abnormalities in podocytes and reducing ROS production in glomeruli. In cultured podocytes, acute-phase MR activation via aldosterone stimulation resulted in a rapid increase in intracellular Ca^{2+} concentration. This effect was suppressed by preincubation with finerenone or the TRPC5 inhibitor AC1903. Moreover, 24-hour aldosterone stimulation induced dose-dependent upregulation of TRPC5 expression. This chronic stimulation also triggered enhanced ROS production and actin cytoskeletal remodeling, both of which were also inhibited by finerenone and AC1903. These findings suggest that MR activation

Downloaded from http://ahajournals.org by on February 11, 2026

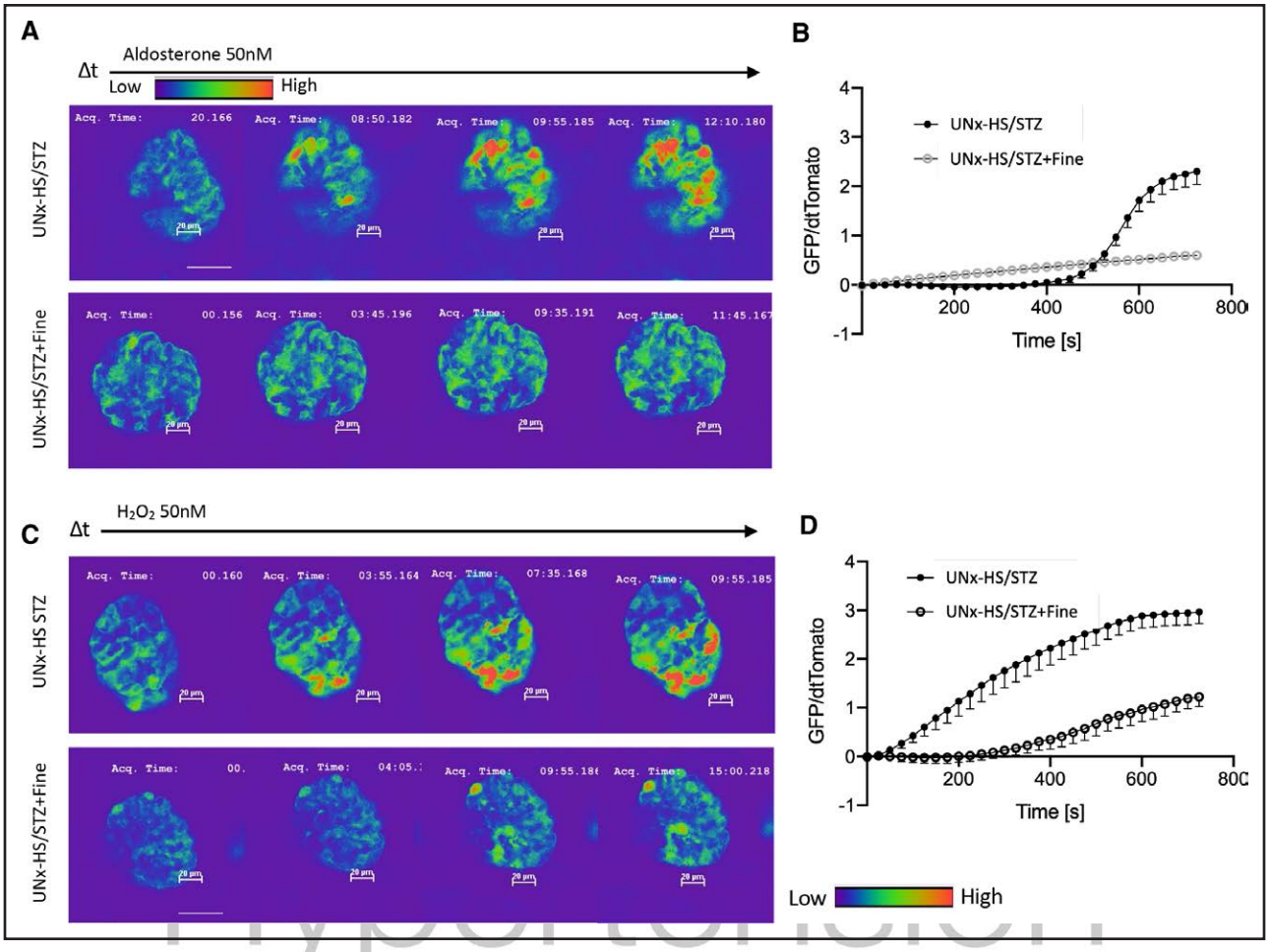

Figure 4. Evaluation of calcium influx into podocytes in isolated glomeruli from streptozotocin (STZ)-induced diabetic podocin-GCaMP5 (genetically encoded calcium indicator 5)/tdTomato (tandem dimer Tomato fluorescent protein) mice following stimulation with aldosterone or H_2O^2.

GCaMP5 fluorescence intensity is shown in the pseudocolor (gradient) map image. **A**, Representative traces of calcium influx imaging in response to aldosterone stimulation in podocytes from the control and finerenone-treated group. **B**, Transient calcium fluxes induced by aldosterone in the absence or presence of finerenone treatment. **C**, Representative traces of calcium influx imaging in response to H_2O_2 stimulation in podocytes from the control and finerenone-treated group. **D**, Transient calcium fluxes induced by H_2O_2 in the absence or presence of finerenone treatment. Scale bar, 40 μm. For isolated glomerulus imaging, a total of 12 to 15 podocytes were analyzed from 3 glomeruli obtained from 2 to 3 mice per group.

Hypertension
FIRST PROOF ONLY

affects podocyte calcium dynamics through both nongenomic and genomic pathways: acutely promoting TRPC channel activation via rapid ROS generation and chronically enhancing TRPC5 expression to promote sustained calcium influx. Finerenone effectively suppressed both of these pathways. In addition, our study demonstrated that finerenone suppressed glomerular hyperfiltration in diabetic mouse models, as evidenced by in vivo imaging-based measurement of SNGFR. The reduction in albuminuria attributed to finerenone is believed to result from both the attenuation of podocyte injury and the inhibition of glomerular hyperfiltration.

We selected the UNx-HS diabetic model because it amplifies diabetic kidney injury within the endogenous diabetic milieu: reduced nephron mass combined with increased distal NaCl delivery increases single-nephron workload, accentuates albuminuria and podocyte stress, and engages diabetes-related Rac1-MR signaling.[18] This pathophysiological context is translationally relevant to patients with diabetes, who often have nephron loss and HS intake. Prior work has successfully used UNx-HS diabetic models to evaluate MR antagonists. In UNx-HS leptin receptor–deficient diabetic mouse model, finerenone reduced albuminuria and glomerular injury, mechanistically linked to suppression of Rac1-MR signaling. In addition, in uninephrectomized leptin receptor–deficient diabetic mouse model without HS feeding, the steroidal MRA eplerenone conferred renoprotection, lower albuminuria, less podocyte injury, and reduced fibrosis,[19] supporting the broader UNx-diabetes framework used to interrogate MR signaling.

In controls, daytime tail-cuff SBP did not differ significantly between the normal-salt and HS diets although values tended to be lower with HS at both 4 and 8

Downloaded from http://ahajournals.org by on February 11, 2026

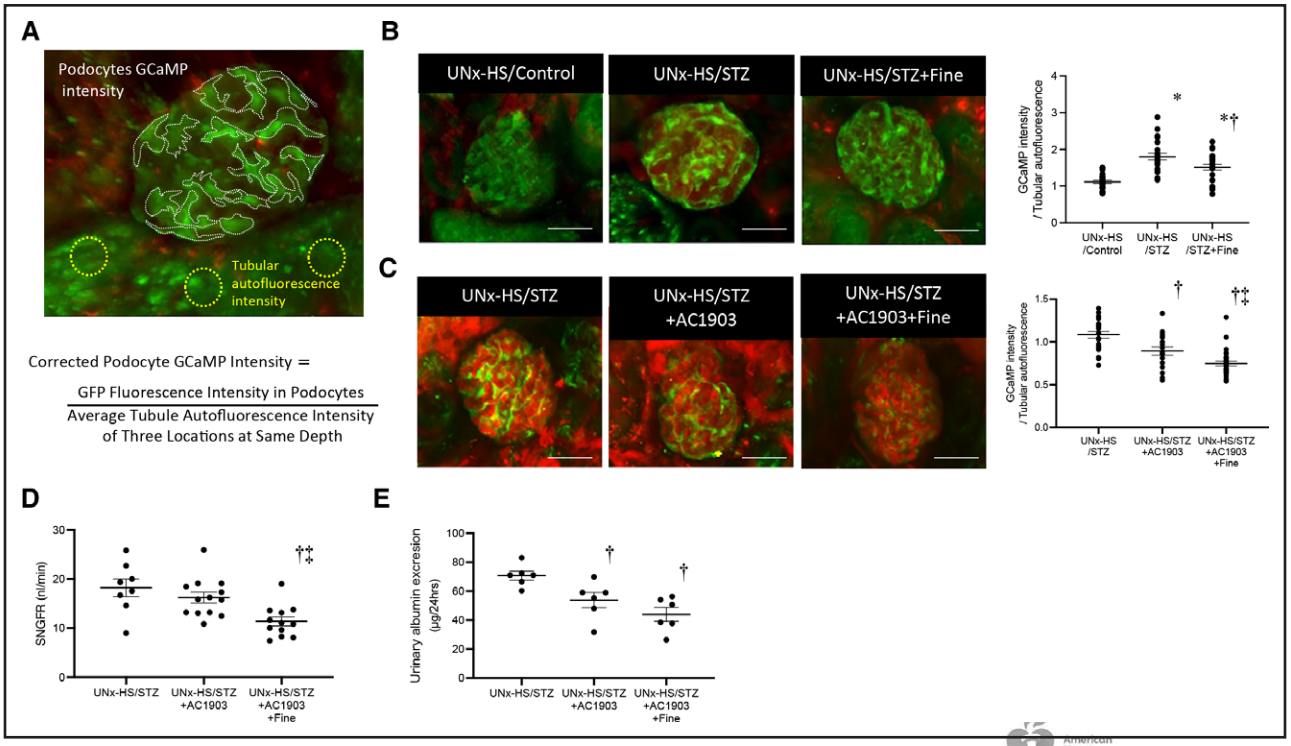

Figure 5. Suppression of podocyte calcium increase in streptozotocin (STZ)-induced diabetic podocin-GCaMP5 (genetically encoded calcium indicator 5)/tdTomato (tandem dimer Tomato fluorescent protein) mice by finerenone and AC1903 (transient receptor potential canonical 5 inhibitor).

A, Schematic representation of the method used to correct and calculate GCaMP5 (GFP [green fluorescent protein]) fluorescence intensity in podocytes. The average autofluorescence intensity of 3 arbitrary locations at the same depth in the tubules was used for correction. **B**, GCaMP5 fluorescence intensity in podocytes of STZ-induced diabetic podocin-GCaMP5/tdTomato mice treated with finerenone for 6 weeks (3–4 mice per group were analyzed, with 5–7 glomeruli per mouse, and a total of 23–26 podocytes per group were measured). **C**, GCaMP5 fluorescence intensity in podocytes of STZ-induced diabetic podocin-GCaMP5/tdTomato mice treated with AC1903 alone and in combination with finerenone for 6 weeks (3–4 mice per group were analyzed, and 4–6 glomeruli per mouse and a total of 18–25 podocytes per group were measured). **D**, Single-nephron glomerular filtration rate (SNGFR) and (**E**) urinary albumin excretion (UAE) measurements in STZ-induced diabetic podocin-GCaMP5/tdTomato mice treated with AC1903 alone and in combination with finerenone for 6 weeks. Mean±SEM. Scale bar, 40 µm. *$P<0.05$ vs uninephrectomy plus high-salt diet (UNx-HS)/control; †$P<0.05$ vs UNx-HS/STZ; and ‡$P<0.05$ vs UNx-HS/STZ+AC1903.

Hypertension FIRST PROOF ONLY

weeks (Figure S3). Regarding systemic hemodynamics, SBP was assessed under daytime resting tail-cuff conditions, using the mean of 3 consecutive readings at each time point. Prior telemetry studies in C57BL/6J mice indicate that a 3% NaCl diet elicits only a modest, nocturnally confined pressor response with minimal daytime change; thus, under our daytime, resting conditions, a salt-induced rise in SBP may be small or undetectable. Consistent with this, in our data set, the UNx-HS cohorts showed numerically lower SBP at 4 and 8 weeks, without reaching statistical significance. *Ins2*$^{+/Akita}$ mice tended to exhibit lower SBP than controls under the same measurement conditions (Figure S3). This difference can be explained by osmotic diuresis and insulin deficiency–related loss of body mass/lean mass, which together reduce effective circulating volume and cardiac output. Insulin-dependent tubular Na+ reabsorption and sympathetic drive may also be attenuated. Furthermore, adaptive changes after UNx, particularly pressure-natriuresis, could limit systemic pressor responses. Taken together, these factors could plausibly account for the slightly lower SBP observed in *Ins2*$^{+/Akita}$ mice. Importantly, at the dose regimen used in this study, finerenone did not reduce SBP relative to diabetic controls. While renal autoregulation (myogenic response and tubuloglomerular feedback) normally buffers physiological BP fluctuations, though it can be impaired in diabetes, SBP remained unchanged with finerenone in our experiments. Thus, the observed renoprotective effects are unlikely to be mediated by systemic BP lowering and instead align with BP-independent mechanisms involving intrarenal MR-TRPC/oxidative pathways examined here.

MR is present in various tissues, including endothelial cells, vascular smooth muscle cells, cardiomyocytes, and macrophages. In the kidney, MR is widely distributed from glomerular cells to tubular cells, fibroblasts, and inflammatory cells.[20] Upregulation of MR expression is observed in many pathological conditions such as hyperglycemia, obesity, insulin resistance, HS intake, and CKD.[21] This upregulation leads to increased transcription of genes associated with inflammation and fibrosis. Kolkhof et al[22] reported that finerenone administration at

Downloaded from http://ahajournals.org by on February 11, 2026

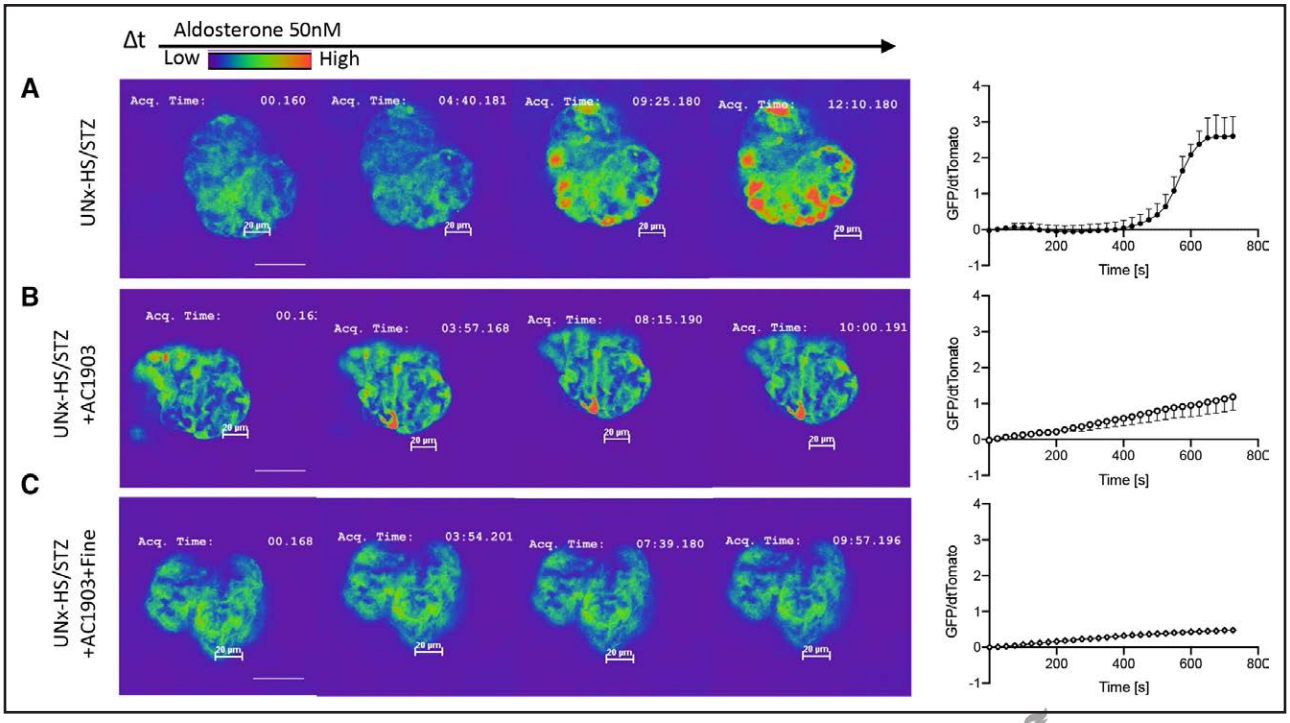

Figure 6. Aldosterone-induced calcium influx into podocytes in isolated glomeruli from streptozotocin (STZ)-induced diabetic podocin-GCaMP5 (genetically encoded calcium indicator 5)/tdTomato (tandem dimer Tomato fluorescent protein) mice treated with AC1903 (transient receptor potential canonical 5 inhibitor) and finerenone.

GCaMP5 fluorescence intensity is shown in the pseudocolor (gradient) map image. Representative traces of calcium influx imaging in response to aldosterone stimulation in podocytes from (**A**) STZ mice, (**B**) STZ mice treated with AC1903, and (**C**) AC1903+finerenone. Scale bar, 40 μm. For isolated glomerulus imaging, a total of 12 to 15 podocytes were analyzed from 3 glomeruli obtained from 2 to 3 mice per group.

Hypertension FIRST PROOF ONLY

doses that did not affect SBP attenuated aldosterone-dependent inflammation, fibrosis, and gene expression of remodeling markers including PAI-1 (plasminogen activator inhibitor-1), osteopontin, KIM-1, MCP-1 (monocyte chemoattractant protein-1), TIMP-1 (tissue inhibitor of metalloproteinases-1), and MMP-2 (matrix metalloproteinase-2) in the renal cortex of deoxycorticosterone acetate-salt hypertensive rats. We have previously reported that MR activation in macrophages induces kidney fibrosis via inflammasome activation mediated by mitochondrial ROS.[23] In addition, MR overactivation is associated with podocyte injury through mechanisms including reduction of autophagy required for podocyte maintenance and increased Nox activity leading to oxidative stress.[24] Based on previous reports indicating that MR activation is involved in the regulation of TRPC expression in coronary smooth muscle cells and cardiomyocytes in animals, we hypothesized that MR activity also contributes to the regulation of TRPC expression and activity in podocytes, and we were able to demonstrate this mechanism in an experimental model of type 1 diabetes.

Because the gain-of-function mutation of TRPC6 was found to cause autosomal dominant focal segmental glomerulosclerosis, TRPC channels have been associated with the progression of various kidney diseases.[25–28] Several genetic types of progressive kidney disease, focal segmental glomerulosclerosis, are caused by mutations in genes encoding modulators of Rac1.[29] Importantly, Rac1 induces TRPC5 activation and cytoskeletal remodeling in podocytes, and the TRPC5 inhibitor AC1903 suppresses kidney disease progression in animal models.[28] Our experiments also showed that increased expression and activity of TRPC5 by MR activation promote calcium influx into podocytes and are associated with podocyte injury. However, there are also previous reports, suggesting that TRPC5 is not associated with podocyte injury or glomerular damage. Overexpression of wild-type TRPC5 or dominant-negative TRPC5 mutant in a C57BL/6 mouse model did not cause albuminuria, foot process effacement, or glomerulosclerosis up to 8 months of age and did not affect LPS-induced albuminuria. Furthermore, administration of the TRPC5 activator englerin A did not induce albuminuria in wild-type mice or mice overexpressing either TRPC5 or a dominant-negative TRPC5 mutant.[30] Originally, the study was not powered or designed to evaluate renal outcomes with AC1903 monotherapy; however, in ancillary analyses of the streptozotocin imaging cohorts, we obtained urinary albumin excretion and SNGFR. AC1903 alone significantly lowered urinary albumin excretion versus diabetic controls while having little effect on SNGFR. In the AC1903+finerenone group, both urinary albumin excretion and SNGFR were significantly reduced; albuminuria

Downloaded from http://ahajournals.org by on February 11, 2026

was numerically lower than with AC1903 monotherapy although the pairwise difference did not reach statistical significance. Taken together, these findings support a model in which TRPC5-dependent mechanisms contribute to podocyte barrier injury/leakiness, whereas attenuation of glomerular hyperfiltration likely reflects TRPC5-independent, MR-linked hemodynamic actions. Moreover, the additivity in Ca^{2+} imaging is consistent with TRPC5-independent pathways, potentially including other TRPC channels.

The podocyte injury induced by MR overactivation demonstrated in this study is a condition that develops chronically and is difficult to explain as a mechanism for the early decrease in albuminuria with finerenone. The study results indicate that finerenone suppresses glomerular hyperfiltration, suggesting that MR activation may control GFR. The impact of aldosterone-induced glomerular microcirculation has been investigated using micropuncture techniques in isolated glomeruli along with the macula densa.[31] Although aldosterone induced vasoconstriction in both the afferent and efferent arterioles, it was shown to cause stronger constriction in the efferent arterioles, potentially increasing glomerular capillary pressure. However, because this effect was not blocked by spironolactone, it was considered to be mediated via MR-independent pathways. Recent studies have shown the presence of MR in macula densa cells, and it has been reported that macula densa cell MRs suppress TG feedback.[32] MR overactivation in the macula densa is also suggested to increase NO production from NOS1 (neuronal nitric oxide synthase) and act as a recruitment factor for NKCC2 (Na-K-2Cl cotransporter 2) although the detailed molecular mechanisms involved are unclear. In addition, increased glucose reabsorption by SGLT1 in macula densa cells has been reported to increase NO production from nNOS (neuronal nitric oxide synthase) and reduce TG feedback.[33] While this study did not provide a detailed examination of the GFR regulatory function associated with MR activation, it is highly likely that this mechanism is important for the improvement of albuminuria with finerenone, and it might be investigated further using biomarker analyses of the CONFIDENCE trial.

This study has several limitations. First, we did not obtain direct evidence demonstrating the involvement of the podocyte-specific MR-TRPC5 pathway. Therefore, future studies should directly investigate the role of the MR-TRPC5 pathway specifically in podocytes. Second, although we evaluated TRPC5 function and expression in this study, we were unable to assess TRPC6. Despite using several commercially available antibodies, TRPC6 expression could not be detected in cultured podocytes. This may be due either to the low expression levels of TRPC6 in cultured cells or to limitations related to antibody quality. Third, cultured podocytes do not fully replicate the morphology of in vivo podocytes. Therefore, it remains uncertain whether the changes in the actin cytoskeleton observed in cultured podocytes also occur in vivo. Fourth, the AC1903 cohorts were originally designed for intravital imaging rather than fully powered outcome analyses; accordingly, sample size is limited, and no formal histopathologic assessment was performed for these add-on groups. Nevertheless, the convergent improvements in albuminuria and podocyte Ca^{2+} dynamics provide in vivo evidence that TRPC5 activity contributes to albuminuric injury, whereas finerenone's antihyperfiltration effect is TRPC5-independent. Fifth, only male *Ins2*$^{+/Akita}$ mice were studied to ensure a consistent diabetic phenotype. On the C57BL/6 background, females typically develop milder and delayed hyperglycemia and albuminuria compared with males, which can blunt disease severity and reduce statistical power; therefore, analyses were restricted to males in the present study. This sex difference is generally attributed to estrogen-mediated β-cell protection, attenuation of endoplasmic reticulum stress and apoptosis in Ins2-mutant β cells, together with greater whole-body insulin sensitivity (and lower hepatic glucose output) in females, resulting in lower glycemia.[34,35]

In conclusion, finerenone ameliorated podocyte injury and albuminuria in diabetic models. These benefits were associated with reduced glomerular ROS and suppression of TRPC5-dependent Ca^{2+} entry in podocytes and likely also involved TRPC5-independent, MR-linked intrarenal hemodynamic mechanisms that mitigate hyperfiltration. Thus, finerenone's renoprotection appears to include multiple pathways rather than rely on a single one. Further works are required to understand the role of these pathways in finerenone-mediated kidney protection in related disease states including type 2 diabetes and nondiabetic CKD.[4,36]

PERSPECTIVES

This study indicates that finerenone ameliorates albuminuria in diabetic kidney disease, plausibly through modulation of podocyte calcium homeostasis and stabilization of the filtration barrier. Beyond the immediate findings, these data reinforce the podocyte as a tractable therapeutic locus for MR-targeted strategies. The observed linkage between MR signaling, calcium dynamics, and cytoskeletal remodeling suggests a broader framework for understanding glomerular hemodynamic-structural coupling. Clinically, these insights may help refine patient selection and end points for trials of MR antagonists, alone or in combination with agents such as SGLT2 inhibitors. Future work should test podocyte-specific MR pathways in vivo, define upstream/downstream effectors (eg, channel and cytoskeletal regulators), and clarify durability and generalizability across diabetes phenotypes and

Hypertension FIRST PROOF ONLY

Downloaded from http://ahajournals.org by on February 11, 2026

comorbid states. If validated, this mechanistic axis could inform precision approaches that align targeted MR modulation with biomarker-guided monitoring to slow kidney disease progression.

ARTICLE INFORMATION

Received July 28, 2025; accepted January 5, 2026.

Affiliations

Department of Nephrology and Hypertension (T.I., K.K., R.T., A.H., E.K., M.T., M.W., Y.W., S.K., H.N., T.S.), Department of General Geriatric Medicine (H.K.), and Kawasaki Geriatric Medical Center (N.K.), Kawasaki Medical School, Okayama, Japan. Division of Nephrology, Department of Medicine, University Health Network, University of Toronto, ON, Canada (D.Z.I.C.).

Acknowledgments

The authors thank Prof Katsuhiko Asanuma (Department of Nephrology, Chiba University Graduate School of Medicine, Chiba, Japan) for providing the cultured podocytes.

Author Contributions

T. Iwakura, K. Kidokoro, and N. Kashihara conceived and designed the study. T. Iwakura, Y. Wada, R. Tatsugawa, A. Hirano, and K. Kidokoro performed experiments and data acquisition. T. Iwakura and K. Kidokoro analyzed and interpreted the data. T. Iwakura, N. Kashihara, and K. Kidokoro drafted the paper. E. Kajimoto, M. Takasu, M. Wada, H. Kadoya, S. Kishi, H. Nagasu, T. Sasaki, D.Z.I. Cherney, and N. Kashihara performed critical revision.

Source of Funding

This investigation was supported by the KAWASAKI Foundation for Medical Science and Medical Welfare, the Research Project Grant from Kawasaki Medical School, and a scholarship donation from Bayer. D.Z.I. Cherney is supported by the Department of Medicine, University of Toronto Merit Award and receives support from the CIHR, the Heart and Stroke Richard Lewar Center of Excellence, the Heart and Stroke Foundation, and the Kidney Foundation of Canada.

Disclosures

This study was funded by Bayer. D.Z.I. Cherney received honoraria from Boehringer Ingelheim-Lilly, Merck, AstraZeneca, Sanofi, Mitsubishi-Tanabe, AbbVie, Janssen, Bayer, Prometic, BMS, Maze, Gilead, CSL-Behring, Otsuka, Novartis, Youngene, Lexicon, Inversago, and Novo Nordisk and received operational funding for clinical trials from Boehringer Ingelheim-Lilly, Merck, Janssen, Sanofi, AstraZeneca, CSL-Behring, and Novo Nordisk. The other authors report no conflicts.

Downloaded from http://ahajournals.org by on February 11, 2026

Hypertension FIRST PROOF ONLY

REFERENCES

1. Bakris GL, Agarwal R, Anker SD, Pitt B, Ruilope LM, Rossing P, Kolkhof P, Nowack C, Schloemer P, Joseph A, et al; FIDELIO-DKD Investigators. Effect of finerenone on chronic kidney disease outcomes in type 2 diabetes. *N Engl J Med*. 2020;383:2219–2229. doi: 10.1056/NEJMoa2025845
2. Pitt B, Filippatos G, Agarwal R, Anker SD, Bakris GL, Rossing P, Joseph A, Kolkhof P, Nowack C, Schloemer P, et al; FIGARO-DKD Investigators. Cardiovascular events with finerenone in kidney disease and type 2 diabetes. *N Engl J Med*. 2021;385:2252–2263. doi: 10.1056/NEJMoa2110956
3. Agarwal R, Filippatos G, Pitt B, Anker SD, Rossing P, Joseph A, Kolkhof P, Nowack C, Gebel M, Ruilope LM, et al; FIDELIO-DKD and FIGARO-DKD investigators. Cardiovascular and kidney outcomes with finerenone in patients with type 2 diabetes and chronic kidney disease: the FIDELITY pooled analysis. *Eur Heart J*. 2022;43:474–484. doi: 10.1093/eurheartj/ehab777
4. Heerspink HJL, Birkenfeld AL, Cherney DZI, Colhoun HM, Ji L, Mathieu C, Groop PH, Pratley RE, Rosas SE, Rossing P, et al. Rationale and design of a randomised phase III registration trial investigating finerenone in participants with type 1 diabetes and chronic kidney disease: the FINE-ONE trial. *Diabetes Res Clin Pract*. 2023;204:110908. doi: 10.1016/j.diabres.2023.110908
5. Nagase M, Fujita T. Endocrinological aspects of proteinuria and podocytopathy in diabetes: role of the aldosterone/mineralocorticoid receptor system. *Curr Diabetes Rev*. 2011;7:8–16. doi: 10.2174/157339911794273919
6. Tian D, Jacobo SM, Billing D, Rozkalne A, Gage SD, Anagnostou T, Pavenstadt H, Hsu HH, Schlondorff J, Ramos A, et al. Antagonistic regulation of actin dynamics and cell motility by TRPC5 and TRPC6 channels. *Sci Signal*. 2010;3:ra77. doi: 10.1126/scisignal.2001200
7. Reiser J, Polu KR, Moller CC, Kenlan P, Altintas MM, Wei C, Faul C, Herbert S, Villegas I, Avila-Casado C, et al. TRPC6 is a glomerular slit diaphragm-associated channel required for normal renal function. *Nat Genet*. 2005;37:739–744. doi: 10.1038/ng1592
8. Eckel J, Lavin PJ, Finch EA, Mukerji N, Burch J, Gbadegesin R, Wu G, Bowling B, Byrd A, Hall G, et al. TRPC6 enhances angiotensin II-induced albuminuria. *J Am Soc Nephrol*. 2011;22:526–535. doi: 10.1681/ASN.2010050522
9. Li W, Chen X, Riley AM, Hiett SC, Temm CJ, Beli E, Long X, Chakraborty S, Alloosh M, White FA, et al. Long-term spironolactone treatment reduces coronary TRPC expression, vasoconstriction, and atherosclerosis in metabolic syndrome pigs. *Basic Res Cardiol*. 2017;112:54. doi: 10.1007/s00395-017-0643-0
10. Iwashima F, Yoshimoto T, Minami I, Sakurada M, Hirono Y, Hirata Y. Aldosterone induces superoxide generation via Rac1 activation in endothelial cells. *Endocrinology*. 2008;149:1009–1014. doi: 10.1210/en.2007-0864
11. Kidokoro K, Cherney DZI, Bozovic A, Nagasu H, Satoh M, Kanda E, Sasaki T, Kashihara N. Evaluation of glomerular hemodynamic function by empagliflozin in diabetic mice using in vivo imaging. *Circulation*. 2019;140:303–315. doi: 10.1161/CIRCULATIONAHA.118.037418
12. Kang JJ, Toma I, Sipos A, McCulloch F, Peti-Peterdi J. Quantitative imaging of basic functions in renal (patho)physiology. *Am J Physiol Renal Physiol*. 2006;291:F495–F502. doi: 10.1152/ajprenal.00521.2005
13. Satoh M, Fujimoto S, Haruna Y, Arakawa S, Horike H, Komai N, Sasaki T, Tsujioka K, Makino H, Kashihara N. NAD(P)H oxidase and uncoupled nitric oxide synthase are major sources of glomerular superoxide in rats with experimental diabetic nephropathy. *Am J Physiol Renal Physiol*. 2005;288:F1144–F1152. doi: 10.1152/ajprenal.00221.2004
14. Nagasu H, Satoh M, Kidokoro K, Nishi Y, Channon KM, Sasaki T, Kashihara N. Endothelial dysfunction promotes the transition from compensatory renal hypertrophy to kidney injury after unilateral nephrectomy in mice. *Am J Physiol Renal Physiol*. 2012;302:F1402–F1408. doi: 10.1152/ajprenal.00459.2011
15. Mundel P, Reiser J, Zuniga Mejia Borja A, Pavenstadt H, Davidson GR, Kriz W, Zeller R. Rearrangements of the cytoskeleton and cell contacts induce process formation during differentiation of conditionally immortalized mouse podocyte cell lines. *Exp Cell Res*. 1997;236:248–258. doi: 10.1006/excr.1997.3739
16. Kidokoro K, Kadoya H, Cherney DZI, Kondo M, Wada Y, Umeno R, Kishi S, Nagasu H, Nagai K, Suzuki T, et al. Insights into the regulation of GFR by the Keap1-Nrf2 pathway. *Kidney360*. 2023;4:1454–1466. doi: 10.34067/KID.0000000000000171
17. Nagasu H, Satoh M, Yorimitsu D, Tomita N, Sasaki T, Kashihara N. Comparison of combination therapy of olmesartan plus azelnidipine or hydrochlorothiazide on renal and vascular damage in SHR/NDmcr-cp rats. *Kidney Blood Press Res*. 2011;34:87–96. doi: 10.1159/000323535
18. Hirohama D, Nishimoto M, Ayuzawa N, Kawarazaki W, Fujii W, Oba S, Shibata S, Marumo T, Fujita T. Activation of Rac1-mineralocorticoid receptor pathway contributes to renal injury in salt-loaded db/db mice. *Hypertension*. 2021;78:82–93. doi: 10.1161/HYPERTENSIONAHA.121.17263
19. Zhou G, Johansson U, Peng XR, Bamberg K, Huang Y. An additive effect of eplerenone to ACE inhibitor on slowing the progression of diabetic nephropathy in the db/db mice. *Am J Transl Res*. 2016;8:1339–1354.
20. Barrera-Chimal J, Lima-Posada I, Bakris GL, Jaisser F. Mineralocorticoid receptor antagonists in diabetic kidney disease - mechanistic and therapeutic effects. *Nat Rev Nephrol*. 2022;18:56–70. doi: 10.1038/s41581-021-00490-8
21. Barrera-Chimal J, Girerd S, Jaisser F. Mineralocorticoid receptor antagonists and kidney diseases: pathophysiological basis. *Kidney Int*. 2019;96:302–319. doi: 10.1016/j.kint.2019.02.030
22. Kolkhof P, Delbeck M, Kretschmer A, Steinke W, Hartmann E, Barfacker L, Eitner F, Albrecht-Kupper B, Schafer S. Finerenone, a novel selective nonsteroidal mineralocorticoid receptor antagonist protects from rat cardiorenal injury. *J Cardiovasc Pharmacol*. 2014;64:69–78. doi: 10.1097/FJC.0000000000000091
23. Kadoya H, Satoh M, Sasaki T, Taniguchi S, Takahashi M, Kashihara N. Excess aldosterone is a critical danger signal for inflammasome activation in the development of renal fibrosis in mice. *FASEB J*. 2015;29:3899–3910. doi: 10.1096/fj.15-271734
24. Yuan Y, Xu X, Zhao C, Zhao M, Wang H, Zhang B, Wang N, Mao H, Zhang A, Xing C. The roles of oxidative stress, endoplasmic reticulum stress, and autophagy in aldosterone/mineralocorticoid receptor-induced podocyte injury. *Lab Invest*. 2015;95:1374–1386. doi: 10.1038/labinvest.2015.118

25. Winn MP, Conlon PJ, Lynn KL, Farrington MK, Creazzo T, Hawkins AF, Daskalakis N, Kwan SY, Ebersviller S, Burchette JL, et al. A mutation in the TRPC6 cation channel causes familial focal segmental glomerulosclerosis. *Science*. 2005;308:1801–1804. doi: 10.1126/science.1106215
26. Ilatovskaya DV, Blass G, Palygin O, Levchenko V, Pavlov TS, Grzybowski MN, Winsor K, Shuyskiy LS, Geurts AM, Cowley AW Jr, et al. A NOX4/TRPC6 pathway in podocyte calcium regulation and renal damage in diabetic kidney disease. *J Am Soc Nephrol*. 2018;29:1917–1927. doi: 10.1681/ASN.2018030280
27. Binz-Lotter J, Jungst C, Rinschen MM, Koehler S, Zentis P, Schauss A, Schermer B, Benzing T, Hackl MJ. Injured podocytes are sensitized to angiotensin II-induced calcium signaling. *J Am Soc Nephrol*. 2020;31:532–542. doi: 10.1681/ASN.2019020109
28. Zhou Y, Castonguay P, Sidhom EH, Clark AR, Dvela-Levitt M, Kim S, Sieber J, Wieder N, Jung JY, Andreeva S, et al. A small-molecule inhibitor of TRPC5 ion channels suppresses progressive kidney disease in animal models. *Science*. 2017;358:1332–1336. doi: 10.1126/science.aal4178
29. Akilesh S, Suleiman H, Yu H, Stander MC, Lavin P, Gbadegesin R, Antignac C, Pollak M, Kopp JB, Winn MP, et al. Arhgap24 inactivates Rac1 in mouse podocytes, and a mutant form is associated with familial focal segmental glomerulosclerosis. *J Clin Invest*. 2011;121:4127–4137. doi: 10.1172/JCI46458
30. Wang X, Dande RR, Yu H, Samelko B, Miller RE, Altintas MM, Reiser J. TRPC5 does not cause or aggravate glomerular disease. *J Am Soc Nephrol*. 2018;29:409–415. doi: 10.1681/ASN.2017060682
31. Arima S, Kohagura K, Xu HL, Sugawara A, Abe T, Satoh F, Takeuchi K, Ito S. Nongenomic vascular action of aldosterone in the glomerular microcirculation. *J Am Soc Nephrol*. 2003;14:2255–2263. doi: 10.1097/01.asn.0000083982.74108.54
32. Fu Y, Hall JE, Lu D, Lin L, Manning RD Jr, Cheng L, Gomez-Sanchez CE, Juncos LA, Liu R. Aldosterone blunts tubuloglomerular feedback by activating macula densa mineralocorticoid receptors. *Hypertension*. 2012;59:599–606. doi: 10.1161/HYPERTENSIONAHA.111.173195
33. Zhang J, Cai J, Cui Y, Jiang S, Wei J, Kim YC, Chan J, Thalakola A, Le T, Xu L, et al. Role of the macula densa sodium glucose cotransporter type 1-neuronal nitric oxide synthase-tubuloglomerular feedback pathway in diabetic hyperfiltration. *Kidney Int*. 2022;101:541–550. doi: 10.1016/j.kint.2021.10.037
34. Gurley SB, Clare SE, Snow KP, Hu A, Meyer TW, Coffman TM. Impact of genetic background on nephropathy in diabetic mice. *Am J Physiol Renal Physiol*. 2006;290:F214–F222. doi: 10.1152/ajprenal.00204.2005
35. Xu B, Allard C, Alvarez-Mercado AI, Fuselier T, Kim JH, Coons LA, Hewitt SC, Urano F, Korach KS, Levin ER, et al. Estrogens promote misfolded proinsulin degradation to protect insulin production and delay diabetes. *Cell Rep*. 2018;24:181–196. doi: 10.1016/j.celrep.2018.06.019
36. Heerspink HJL, Agarwal R, Bakris GL, Cherney DZI, Lam CSP, Neuen BL, Sarafidis PA, Tuttle KR, Wanner C, Brinker MD, et al. Design and baseline characteristics of the finerenone, in addition to standard of care, on the progression of kidney disease in patients with Non-Diabetic Chronic Kidney Disease (FIND-CKD) randomized trial. *Nephrol Dial Transplant*. 2025;40:308. doi: 10.1093/ndt/gfae132

American Heart Association.

Hypertension
FIRST PROOF ONLY

Downloaded from http://ahajournals.org by on February 11, 2026